IMEC online 2022

Category 3 (grades 4 – 5)

1) There are 100 chickens and rabbits on a farm. The total number of legs of the chickens is 80 more than the total number of legs of the rabbits. How many chickens are there?

(a)80 (b)90 (c)100 (d)40

2) The remainder is 3 when a number is divided by 9. The remainder is 2 when the same number is divided by 5. If the number is less than 60, what is the greatest possible such number?

(a)45 (b)33 (c)56 (d)57

3) A4-digit number is formed when Junlin writes her own age right after her father's Age. Their age difference is then subtracted from this number and the result is 3786.How old is Junlin?

(a)13 (b)14 (c)12 (d)15

4) A fish tank has a total of 50 guppies and angelfish. If each guppy is exchanged for 2 angelfish, there will be 86 angelfish altogether. How many guppies are there?

(a)12 (b)32 (c)36 (d)45

5) The cost of 4 packs of nuts, 3 packs of sweets and 2 packs of chocolates is $36.50The cost of 1 pack of nuts and 1 pack of sweets is $6.80. The cost of 1 pack of sweets and 1 pack of chocolates is $8.10. What is the cost of 1 pack of nuts?

(a)$4.50 (b)$5.40 (c)$5.60 (d)3.20

6) Connor has to reach school by 7.15 am every morning. If he travels at 60 meters per minute, he will arrive 8 minutes earlier. If he travels at 50 meters per minute, he will arrive 6 minutes earlier. Given that he leaves home for school after 6.40 am, find the exact time he leaves home for school so that he arrives just on time.

(a)2.57am (b)3.57am (c)4.57am (d)6.57am

7) The length of a rope is 240 cm. From one end of the rope, a marking is made at every 4 cm. From the other end of the rope, a marking is made at every 6 cm. The rope is then cut according to the markings made. How many pieces of rope are there after cutting?

(a)39 (b)89 (c)45 (d)57

8) A number multiplied by itself is a square number. Given that 432 × a is a square number, find the least possible value of a. [Hint: Prime factorize 432]

(a)3 (b)4 (c)6 (d)7

9) 4 books, A, B, C and D, are to be placed into a school bag in such a way that(i)there must be at least one book in the bag;(ii) there can be at most two books in the bag. How many ways are there to do so?

(a)2 (b)9 (c)10 (d)12

10) Each worker in a factory can produce 8 shirts or 14 pairs of pants. One suit consists of 1 shirt and 1 pair of pants. How many suits can 154 workers produce in a day?

(a)784 (b)654 (c)384 (d)574

11) Sam and Ted had $564 altogether. After Ted gave $54 to Sam, Sam had 5 times as much money as Ted. How much money did Ted have at first?

(a)$148 (b)$45 (c)$56 (d)$418

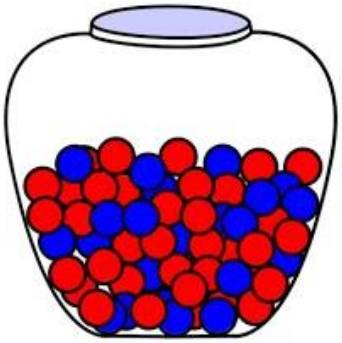
12) A whole number is such that when it is divided by 5, the remainder is 3. The remainder is 2 when the divisors are 3 or 4. The number can be divided by 7. Find the largest of such a 2-digit whole number.

(a)88 (b)98 (c)93 (d)93

13) A farmer placed some eggs into a box in such a way that the number of eggs he placed each time was twice the number of eggs he placed before. He finally filled the box after 8 minutes. At which minute was the box half filled with eggs?

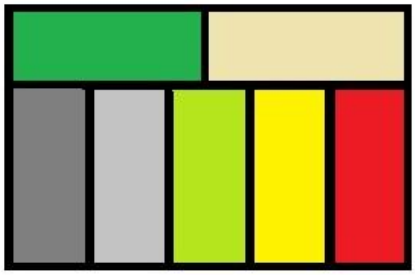
(a)4 (b)6 (c)3 (d)7

14) Bottle A contains 30 blue and red marbles. Bottle B contains some blue and red marbles. How many red marbles must be transferred from Bottle B to Bottle A so that the number of red marbles in Bottle A is the number of marbles in Bottle A?



(a)13 (b)15 (c)24 (d)12

15) A rectangle of perimeter 68 cm is divided into 7 identical rectangles as shown. What is the area of each rectangle?



(a)280 (b)230 (c)250 (d)220

16) 2000-1997+1994-1991+…. +14-11+8-5+4

(a)1001 (b)1002 (c)1003 (d)1004

17) A, B, C are all prime numbers.

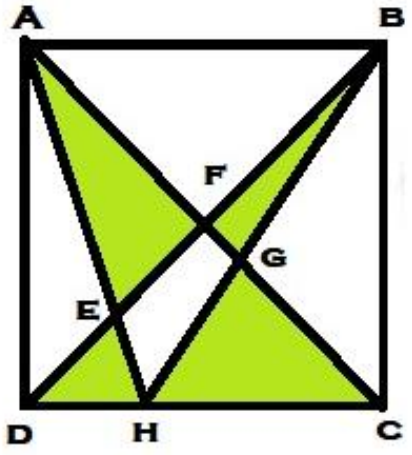
Given A+B=25
B+C=36
C+D=42

Find the value of d.

(a)28 (b)27 (c)26 (d)29

18) In the square shown, the area of quadrilateral EFGH is 12 cm square. Given that the side of the square is 12 cm, find the area of the shaded region.

(a)45 (b)48 (c)46 (d)47



19) 255 numbers beginning from 1,2, 3,…,255 are used for the page numbers of a book. How many times does the number'2' appear?

(a)110 (b)111 (c)122 (d)112

20) $\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\frac{1}{64}=?$

(a) $\frac{60}{64}$ (b) $\frac{62}{64}$ (c) $\frac{63}{64}$ (d) $\frac{61}{64}$

21) A farmer prepared a pile of grass for his herd. The pile of grass was enough to feed 3 cows and 6 goats for 15 days, or 4cows and 10 days. For how many days could the pile of grass feed 4cows and 4 goats.

(a)12 (b)13 (c)14 (d)15

22) (12345678 + 23456781 + 34567812 +……+81234567) ÷ 6.

(a)66 (b)666 (c)6666 (d)66666666

23) 152 children attended a Math Camp. 1/3 the number of boys was equal to 4/7 the number of girls. How many girls attended the math Camp?

(a)34 (b)45 (c)56 (d)67

24) 1,1,3,5,11,21,43,85,171,341,683, ____

(a)1234(b)1365(c)1345(d)1325

25) Evaluate {.} in ({} × 35 -16÷ 0.5) ÷ 8 – 30=1

(a)6(b)7(c)8(d)9

26) 50 students took part in 50-m freestyle and 50-m backstroke swimming events.32 students took part in freestyle and 28 students took part in backstroke. How many students took part in backstroke only?

27) Each letter in the multiplication working below represents a number. Find the multiplication working.

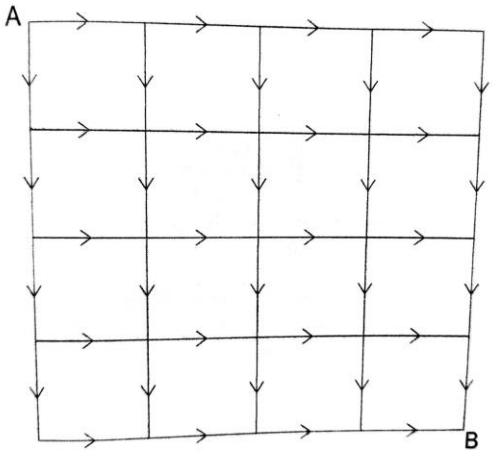
$$\begin{array}{cccc} a & b & c & d \\ & & \times & 9 \\ \hline d & c & b & a \end{array}$$

28) Mrs. Tara paid a bill of $61 with $10, $5 and $2 notes. She used 10 pieces of notes altogether. How many pieces of $10 notes did she use?

29) A, B and C are 3 whole numbers.

Given A+B+C = 6000

B-A= 145

If A 1354, what is the value of (C-A)?

30) How many ways are there to travel from Point A to Point B using the directions shown by the arrows?